

Gukyeong Kwon

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SUMMARY

I am a Senior Applied Scientist at Amazon Artificial General Intelligence (AGI) working on developing multimodal foundation models. In Amazon, I have contributed to the launch of Amazon Titan Multimodal Embeddings model, Amazon Nova family of models, and Amazon Nova Multimodal Embeddings model.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Electrical and Computer Engineering (Advisor: Dr. Ghassan AlRegib)

August 2015 – August 2021

M.S. in Electrical and Computer Engineering (GPA: 4.0/4.0)

August 2015 – May 2018

Sungkyunkwan University (SKKU)

Suwon, South Korea

B.S. in Electronic and Electrical Engineering (GPA: 4.29/4.5)

March 2009 – August 2015

WORK EXPERIENCES

Senior Applied Scientist

Amazon AGI, New York, NY

• Amazon Nova Multimodal Embeddings

January 2025 – November 2025

- Defined the product requirements for the embedding model such as modality support, model size, latency, and target benchmarks through close collaborations with product managers and customers.
- Investigated training strategies for effective contrastive learning, which includes native resolution support, scaling of batch size, hard negative mining, distillation, and instruction tuning.
- Performed data mixture optimization over five different modality data (image, text, document, video, audio) to achieve leading accuracy in the retrieval tasks for all the modalities.
- Contributed to the product API design by providing science guidance on model capabilities and user experiences.
- Led a group of 10 Applied Scientists and Research Engineers by establishing the science plans and prioritization.

• Amazon Nova Multimodal Foundation Model

January 2024 – December 2024

- Developed compute-efficient vision encoder training methods by optimizing model size and image resolution.
- Investigated token merging to compress the number of output tokens while minimizing performance degradation.
- Implemented image representation techniques to support diverse image resolutions and aspect ratios.
- Optimized data mixture over different types of multimodal data for the pre-training and mid-training.
- Conducted benchmark analysis to characterize failure cases of the foundation model, understand the performance bottleneck in specific benchmark domains, and developed targeted solutions to improve the performance.

Applied Scientist

AWS AI Labs, New York, NY

• Amazon Titan Multimodal Embeddings

January 2022 – October 2023

- Prepared billion-scale web data and performed data filtering / enrichment through synthetic caption generation.
- Designed the model architecture by ablating the scaling effect of the vision encoder and the text encoder.
- Built training and evaluation codebases for FLOPS-optimized training and benchmarking across diverse tasks.
- Developed a technique to support multiple embedding sizes in a single model (e.g. 1024, 384, and 256 dimensional embeddings) for customers to have an option to trade-off between cost and performance.

• Vision and Language Representation Learning

January 2021 – December 2022

- Developed a vision and language representation learning method (named MaskVLM), where the masked signal of one modality is reconstructed with the help from another modality.
- Achieved state-of-the-art performance on diverse vision and language tasks such as image-text retrieval, visual question answering, natural language for visual reasoning, and visual entailment.
- Derived theoretical interpretation on the advantages of MaskVLM from a perspective of variation of information.
- Conducted evaluation in limited pre-training data scenarios and highlighted the data-efficiency of MaskVLM.

Graduate Research Assistant

Georgia Tech, Atlanta, GA

• Anomaly Detection Using Gradient-Based Representations

August 2018 – August 2021

- Developed an anomaly detection algorithm using gradient-based representations which characterize knowledge that the model has not learned and achieved state-of-the-art performance in diverse image recognition datasets.
- Investigated an accident event detection algorithm to detect abnormal situations in driving scenarios such as a pedestrian jumping in front of a car or a bumper of car in the middle of the road.
- Incorporated out-of-distribution detection into Faster-RCNN to detect abnormal objects on the road.

PUBLICATIONS

- Amazon Artificial General Intelligence, “Amazon Nova Multimodal Embeddings: Technical report and model card,” *Amazon Science*, 2025. [\[Link\]](#)
- Amazon Artificial General Intelligence, “The Amazon Nova Family of Models: Technical Report and Model Card,” *Amazon Science*, 2024. [\[arXiv\]](#)
- G. Kwon, Z. Cai, A. Ravichandran, E. Bas, R. Bhotika, and S. Soatto, “Masked Vision and Language Modeling for Multi-modal Representation Learning,” *International Conference on Learning Representations (ICLR)*, 2023. [\[arXiv\]](#)
- X. Fu, S. Zhang, G. Kwon, P. Perera, H. Zhu, Y. Zhang, A. Li, W. Wang, Z. Wang, V. Castelli, P. Ng, D. Roth, B. Xiang, “Generate then Select: Open-ended Visual Question Answering Guided by World Knowledge,” *The 61st Annual Meeting of the Association for Computational Linguistics (ACL) Findings*, 2023. [\[arXiv\]](#)
- Z. Cai, G. Kwon, A. Ravichandran, E. Bas, Z. Tu, R. Bhotika, and S. Soatto, “X-DETR: A Versatile Architecture for Instance-wise Vision-Language Task,” *European Conference on Computer Vision (ECCV)*, 2022. [\[arXiv\]](#)
- G. Kwon and G. AlRegib, “A Gating Model for Bias Calibration in Generalized Zero-shot Learning,” *IEEE Transactions on Image Processing*, 2022. [\[arXiv\]](#) [\[GitHub\]](#)
- G. Kwon, M. Prabhushankar, D. Temel and G. AlRegib, “Backpropagated Gradient Representations for Anomaly Detection,” In *European Conference on Computer Vision (ECCV)*, 2020. [\[arXiv\]](#) [\[GitHub\]](#) [\[Video\]](#) [\[Slides\]](#)
- G. Kwon, M. Prabhushankar, D. Temel and G. AlRegib, “Novelty Detection Through Model-based Characterization of Neural Networks,” *2020 IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, United Arab Emirates (UAE), 2020. [\[arXiv\]](#) [\[GitHub\]](#) [\[Video\]](#) [\[Slides\]](#)
- M. Prabhushankar, G. Kwon, D. Temel and G. AlRegib, “Contrastive Explanations in Neural Networks,” *2020 IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, United Arab Emirates (UAE), 2020. (**Top Viewed Special Session Paper Award**) [\[arXiv\]](#) [\[GitHub\]](#) [\[Slides\]](#) [\[Award\]](#)
- G. Kwon*, M. Prabhushankar*, D. Temel and G. AlRegib, “Distorted Representation Space Characterization Through Backpropagated Gradients,” *2019 IEEE International Conference on Image Processing (ICIP)*, Taipei, Taiwan, 2019. (*: equal contribution, **Best Paper Award (top 0.1%)**) [\[arXiv\]](#) [\[GitHub\]](#) [\[Poster\]](#) [\[Award\]](#)
- M. Prabhushankar*, G. Kwon*, D. Temel and G. AlRegib, “Semantically Interpretable and Controllable Filter Sets,” *2018 IEEE International Conference on Image Processing (ICIP)*, Athens, Greece, October 2018. (*: equal contribution) [\[arXiv\]](#) [\[GitHub\]](#) [\[Poster\]](#)
- D. Temel, G. Kwon*, M. Prabhushankar*, and G. AlRegib, “CURE-TSR: Challenging Unreal and Real Environments for Traffic Sign Recognition,” *MLITS workshop in Neural Information Processing Systems (NIPS)*, Long Beach, U.S.A, 2017. (*: equal contribution) [\[arXiv\]](#) [\[GitHub\]](#) [\[Poster\]](#)
- M. Aabed, G. Kwon, and G. AlRegib, “Power of Tempospatially Unified Spectral Density for Perceptual Video Quality Assessment,” *2017 IEEE International Conference on Multimedia and Expo (ICME)*, Hong Kong, 2017. (**Finalist of the World’s FIRST 10K Best Paper Award (top 3%)**) [\[arXiv\]](#) [\[GitHub\]](#) [\[Slides\]](#) [\[Award\]](#)

PATENTS

Non-provisional Patent

- A. Parchami, G. AlRegib, D. Temel, M. Prabhushankar, and G. Kwon, “Variance of gradient based active learning framework for training perception algorithms,” US 12,079,738 B2. [\[Link\]](#)
- G. AlRegib, G. Kwon, M. Prabhushankar, and D. Temel, “Detecting and Classifying Anomalies in Artificial Intelligence Systems,” US 2022/0327389 A1. [\[Link\]](#)

Provisional Patents

- G. AlRegib, M. Prabhushankar, G. Kwon, and D. Temel, “System and Method for Detecting and Protecting Against Deceptive Inputs for AI Systems and Measuring Vulnerabilities of Existing AI Systems,” No: 62/899,783, granted on September 13, 2019.
- G. AlRegib, G. Kwon, M. Prabhushankar, and D. Temel, “Task-Generalizable System and Method for Detecting and Classifying Anomalies in Artificial Intelligence Systems,” No: 62/895,556, granted on September 4, 2019.

PROFESSIONAL SERVICES

- Area Chair for the European Conference on Computer Vision (ECCV).
- Reviewer for CVPR, Transactions on Image Processing, Elsevier Signal processing: Image Communication, ICIP, and ICASSP (Recognized as an Outstanding reviewer in CVPR 2023 and 2025).